

Project Module 1: Fairness Audit and Bias Mitigation **

Format:

Technical report including visualizations + completed jupyter notebook

Overview:

In this project, you will perform a fairness audit of a machine learning model, apply bias mitigation techniques, and evaluate the resulting trade-offs between fairness and accuracy.

You will:

1. Train a baseline model
2. Measure multiple fairness metrics
3. Apply mitigation techniques
4. Compare models
5. Provide deployment recommendations

Dataset Options:

Option 1: German Credit Dataset

Task: Predict creditworthiness

Potential protected attributes: Age, Gender

Typical fairness concerns include loan approval disparities across demographic groups.

Option 2: Law School Admissions Council (LSAC)

Task: Predict bar exam passage or admission outcomes.

Potential protected attributes: Race, Gender

Typical fairness concerns include bias in admissions or exam predictions across demographic groups.

** You are encouraged to use Large Language Models (LLMs) to support your thinking and learning. If you do so, you must include a **public link to the conversation(s)** used.

AUC's academic integrity rules apply: submitted work must reflect your own understanding.

You are provided with two starter jupyter notebooks each containing the loading and the typical data preprocessing of one of the above datasets. Provided code blocks are to be used as is.

Select the dataset you'll use and use its starter notebook to load and preprocess the data. Any code you write should be placed after the already provided code without modifying it.

Project Tasks:

Part 1: Fairness Audit (25%)

1. Train a baseline model of your choice (i.e. logistic regression, decision tree, gradient boosting, etc...) and justify your choice
2. Evaluate standard performance metrics reporting on accuracy, precision, recall and F1-score. Explain what these metrics indicate about your model performance
3. Evaluate fairness metrics such as demographic parity, equalized odds and calibration. You must compute these metrics for each demographic group separately
4. Analyze the results addressing which metrics are violated, which group experiences the largest disparities and how large are these differences
5. Visualize the disparities including group prediction rates, error rate comparisons and fairness metric comparisons. Visualizations must clearly show differences between demographic groups

Part 2: Preprocessing Bias Mitigation (25%)

You will attempt to mitigate bias by modifying the **training data**.

Implement one preprocessing mitigation technique of your choice:

1. Reweighting training samples
2. Resampling demographic groups

Following applying the preprocessing method, retrain the model. Recompute all fairness metrics and compare the results with the baseline model.

Address whether fairness improved, which metrics improved the most, whether the accuracy changed and why would this technique affect fairness

Part 3: In-processing Bias Mitigation (25%)

You will implement a **training-time fairness intervention**.

Examples include:

1. Fairness constraints in loss function
2. Fairness-aware optimization
3. Adversarial debiasing
4. Fairness-aware ML libraries

Evaluate all fairness metrics again and compare the results with the baseline model and the preprocessing mitigation model.

Address which mitigation approach improved fairness most, which metric remained hardest to satisfy and whether the model experienced an accuracy drop

Part 4: Analysis and Recommendations (25%)

Analyze the **trade-offs between fairness and accuracy**.

1. Visualize fairness vs accuracy (i.e. accuracy vs demographic parity, accuracy vs equalized odds gap etc...)
2. Compare the baseline model, preprocessing and in-processing mitigations.
Which model provides the best balance
3. Discuss which fairness metrics should be prioritized? Which group was affected the most? Would you recommend real-life deployment of this model? What real-world harms could arise from biased predictions?
4. Discuss potential limitations of your analysis